

TWIZ: The Multimodal Conversational Task Wizard

Rafael Ferreira
Diogo Silva
Diogo Tavares
Frederico Vicente
rah.ferreira@campus.fct.unl.pt
dmgc.silva@campus.fct.unl.pt
dc.tavares@campus.fct.unl.pt
fm.vicente@campus.fct.unl.pt
Universidade NOVA de Lisboa
Lisbon, Portugal

Mariana Bonito
Gustavo Gonçalves
Rui Margarido
Paula Figueiredo
m.bonito@campus.fct.unl.pt
gs.goncalves@campus.fct.unl.pt
rp.margarido@campus.fct.unl.pt
pc.mestre@campus.fct.unl.pt
Universidade NOVA de Lisboa
Lisbon, Portugal

Helder Rodrigues
David Semedo
Joao Magalhaes
harr@campus.fct.unl.pt
df.semedo@fct.unl.pt
jm.magalhaes@fct.unl.pt
Universidade NOVA de Lisboa
Lisbon, Portugal

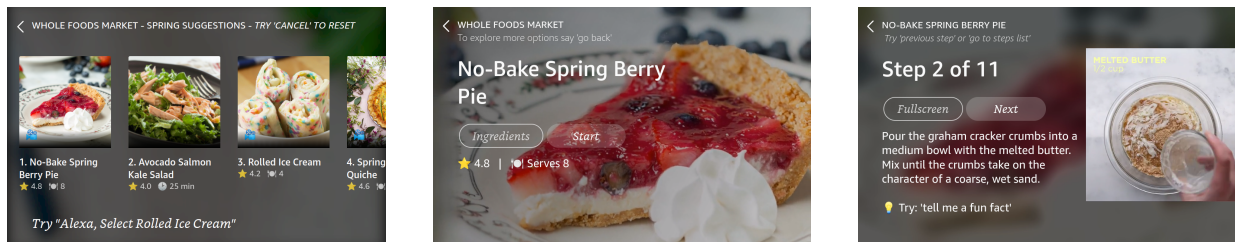


Figure 1: (a) Task Grounding; (b) Task Overview; (c) Step Instructions.

ABSTRACT

This paper introduces TWIZ, a multimodal conversational task wizard that supports an engaging experience, where users are guided through a multimodal conversation, towards the successful completion of recipes and DIY tasks. TWIZ leverages task guides from WikiHow and Recipe sources, as well as dialog AI-based methods to deliver a rich, compelling, and engaging experience when guiding users through complex manual tasks. TWIZ participated in the Amazon Alexa Prize Taskbot 2021 [3]. [Demo Video](#).

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing: Intelligent agents**; • **Human-centered computing** → **Interactive systems and tools**.

KEYWORDS

Conversation Assistants, Multimodal, Artificial Intelligence, Natural Language Processing

ACM Reference Format:

Rafael Ferreira, Diogo Silva, Diogo Tavares, Frederico Vicente, Mariana Bonito, Gustavo Gonçalves, Rui Margarido, Paula Figueiredo, Helder Rodrigues, David Semedo, and Joao Magalhaes. 2022. TWIZ: The Multimodal Conversational Task Wizard. In *Proceedings of the 30th ACM International*

Conference on Multimedia (MM '22), October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3503161.3547741>

1 MULTIMODAL TASK GUIDING AGENTS

Research in multimodal conversational AI is currently tackling the problem of complex manual tasks such as Cooking and DIY. In this novel setting, task complexity needs to be managed and delivered in a controlled manner to effectively assist the user. The core idea is to guide users in an engaging manner by distributing cognitive load throughout the task, thus seeking a symbiosis between task completion and a multimodal curiosity-exploration paradigm. The key novelties and application design principles are:

Conversational task-grounding. Robust intent detection and retrieval component, tailored for voice-based conversational agents, to quickly find the desired task and minimize user frustration - Figure 1 (a);

Task organization and presentation. Task steps are decomposed and automatically illustrated with images, thus improving clarity and providing instructional visual cues. When available, we use videos to complement task instructions - Figure 1 (c);

Conversational engagement. Novel features aimed at keeping users engaged throughout the conversation: 3D tasks visual preview, task-specific curiosities through dialog system initiative, and support for contextual question-answering.

2 CONVERSATIONAL SYSTEM DESIGN

Our architecture is presented in Figure 2. It is based on the CoBot framework [7] provided by Amazon. This framework runs the main flow of the agent on a server-less cloud function, and uses a database to save the dialog turns and user state. The ML algorithms are run

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
MM '22, October 10–14, 2022, Lisboa, Portugal
© 2022 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9203-7/22/10.
<https://doi.org/10.1145/3503161.3547741>

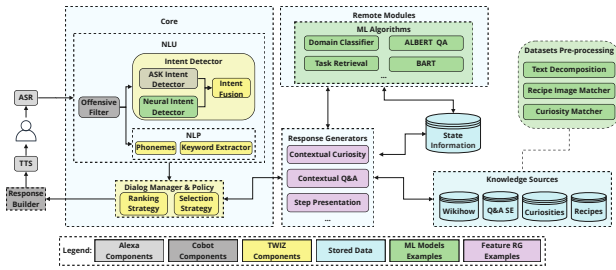


Figure 2: TWIZ's components architecture.

on docker containers in remote computing platforms, reducing the load on the cloud function.

2.1 Multimodal Knowledge Bases and Tasks

Interested in providing meaningful interactions with users, we put effort into collecting and curating custom knowledge bases. First, it is relevant to mention that TWIZ is bounded to two task domains: DIY and recipes. Therefore, in the DIY domain, we used articles from WikiHow¹, and with respect to recipes, we used an API provided by Amazon composed of several recipe sources. Secondly, it was our interest to create dynamic and interactive experiences, for which we curated a contextualized curiosity knowledge base (e.g. "There is a wild mushroom that tastes nearly identical to chicken"). Users also often feel the need to be clarified over the domain of the task at hand. Therefore to answer users' questions, we assembled a collective dataset of QAs forums related to the cooking and DIY domains (ELI5, AskScience [2], and StackExchange² forums). With these datasets, we trained a BART model [9] on a QA task in order to provide domain-specific natural/human-like responses.

2.2 Robust Dialog Interaction

To keep a robust dialog interaction, we developed a dialog manager that orchestrates the multiple components of the system (Figure 2), towards keeping the flow of the conversation. As a result, it is composed of 5 major stages:

1. *Start* - Greets the user and suggests common use cases.
2. *Task Grounding/Search* - After the user issues a search request, we present the results retrieved by our search pipeline (Section 2.2.2).
3. *Task Overview* - After the user selects a task. We show a brief task overview to ensure the task conforms to the user's criteria.
4. *Step-by-Step* - The user starts the task and engages with the multimodal engagement system components (Section 2.3).
5. *Task Concluded* - At the end of the task, we thank the user for the interaction and end with a contextual message.

2.2.1 Intent Detection. To understand the user's requests, we use an ensemble of intent detection models. In specific, we use three different methods: (1) phonemes-based matching using [11] to improve robustness to ASR errors, (2) a BERT-based model for intent detection [1] trained on a manually annotated set of in-domain

conversations, and (3) the intent detection model provided by the Alexa's Skill Kit³. We then fuse these intents based on specific thresholds experimentally set by replaying previous dialogs. Finally, we apply rule-based corrections for outlier situations.

2.2.2 Search Pipeline. A crucial component for system performance is to search for relevant tasks. To be more robust, we used a combination of BM25 and a DPR model [6] trained with queries generated from titles using [13]. Lastly, we rank the search results' titles via the Jaccard similarity against the query.

2.3 Multimodal Engagement

One of our central motivations was to guarantee an enjoyable interaction with TWIZ, and improve the overall user experience through the creation of well-paced and engaging interaction flows.

2.3.1 Task Decomposition. To keep the instructions simple and manageable, we decompose long steps into multiple steps, using [4] to detect sentences and breaking each step after a certain word threshold has been reached.

2.3.2 Contextual QA. Additionally, we also provide support for contextual QA over the task's steps using an ALBERT [8] model fine-tuned on [14], and provide ingredient replacements using [15].

2.3.3 Dynamic 3D KenBurns Overview. To persuade the user to select a recipe, we provide dynamic content in the form of a 3D KenBurns Effect [10] over the recipe's image, thus providing a 3D Zoom out effect as it can be observed in Figure 1 (b).

2.3.4 Step Videos. TWIZ enriches the step instructions with a video illustrating the step text (Figure 1 (c)). These videos were extracted from both recipes and WikiHow articles.

2.3.5 Automated Recipe Illustration. In the recipes, we noticed that a significant portion did not have an image associated with each step. Therefore, we use the recipe's information, and apply a weighted *TF-IDF* [5] to obtain the top-10 similar step-images which are then re-ranked using the score given by CLIP [12].

3 CONCLUSIONS AND IMPACT

In this paper, we presented our approach to the current challenges faced by the rise of voice assistants' popularity. As this topic gains more attention, so does the complexity of the problems it must address. TWIZ aims to tackle the complexity issue in task-oriented voice assistants by managing the user's cognitive load during the task while providing an engaging curiosity-exploration paradigm. This demo will showcase how we successfully addressed these challenges with state-of-the-art concepts and algorithms.

ACKNOWLEDGMENTS

This work has been partially funded by the Amazon Science - TaskBot prize Challenge, and by the NOVA LINC project Ref. UIDP/04516/2020. We also gratefully acknowledge the support of the NVIDIA Corporation regarding the GPUs used for this research.

¹<https://www.wikihow.com/>

²<https://archive.org/download/stackexchange>

³<https://developer.amazon.com/en-US/alexa/alexa-skills-kit>

REFERENCES

- [1] Guan-Lin Chao and Ian R. Lane. 2019. BERT-DST: Scalable End-to-End Dialogue State Tracking with Bidirectional Encoder Representations from Transformer. In *Interspeech 2019, 20th Annual Conference of the International Speech Communication Association, Graz, Austria, 15-19 September 2019*, Gernot Kubin and Zdravko Kacic (Eds.). ISCA, 1468–1472. <https://doi.org/10.21437/Interspeech.2019-1355>
- [2] Angela Fan, Yacine Jernite, Ethan Perez, David Grangier, Jason Weston, and Michael Auli. 2019. ELI5: Long Form Question Answering. In *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, Anna Korhonen, David R. Traum, and Lluís Màrquez (Eds.). Association for Computational Linguistics, 3558–3567. <https://doi.org/10.18653/v1/p19-1346>
- [3] Rafael Ferreira, Diogo Silva, Diogo Tavares, Frederico Vicente, Mariana Bonito, Gustavo Goncalves, Rui Margarido, Paula Figueiredo, Helder Rodrigues, David Semedo, and Joao Magalhaes. 2022. TWIZ: A conversational Task Wizard with multimodal curiosity-exploration. In *Alexa Prize TaskBot Challenge Proceedings*. <https://www.amazon.science/alexa-prize/proceedings/twiz-a-conversational-task-wizard-with-multimodal-curiosity-exploration>
- [4] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. spaCy: Industrial-strength Natural Language Processing in Python. (2020). <https://doi.org/10.5281/zenodo.1212303>
- [5] Karen Spärck Jones. 2004. A statistical interpretation of term specificity and its application in retrieval. *J. Documentation* 60, 5 (2004), 493–502. <https://doi.org/10.1108/00220410410560573>
- [6] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick S. H. Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020*, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, 6769–6781. <https://doi.org/10.18653/v1/2020.emnlp-main.550>
- [7] Chandra Khatrri, Anu Venkatesh, Behnam Hedayatnia, Raefer Gabriel, Ashwin Ram, and Rohit Prasad. 2018. Alexa Prize — State of the Art in Conversational AI. *AI Magazine* 39, 3 (Sep. 2018), 40–55. <https://doi.org/10.1609/aimag.v39i3.2810>
- [8] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. 2019. ALBERT: A Lite BERT for Self-supervised Learning of Language Representations. <https://doi.org/10.48550/ARXIV.1909.11942>
- [9] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, Online, 7871–7880. <https://doi.org/10.18653/v1/2020.acl-main.703>
- [10] Simon Niklaus, Long Mai, Jimei Yang, and Feng Liu. 2019. 3D Ken Burns Effect from a Single Image. *ACM Transactions on Graphics* 38, 6 (2019), 184:1–184:15.
- [11] Jongseok Park, Kyubyong & Kim. 2019. g2pE. <https://github.com/Kyubyong/g2p>.
- [12] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*, Marina Meila and Tong Zhang (Eds.). PMLR, 8748–8763. <https://proceedings.mlr.press/v139/radford21a.html>
- [13] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *J. Mach. Learn. Res.* 21 (2020), 140:1–140:67. <http://jmlr.org/papers/v21/20-074.html>
- [14] Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know What You Don't Know: Unanswerable Questions for SQuAD. <https://doi.org/10.48550/ARXIV.1806.03822>
- [15] Sola Shirai, Oshani Seneviratne, Minor Gordon, Ching-Hua Chen, and Deborah L. McGuinness. 2020. Semantics-Driven Ingredient Substitution in the FoodKG. In *Proceedings of the ISWC 2020 Demos and Industry Tracks: From Novel Ideas to Industrial Practice co-located with 19th International Semantic Web Conference (ISWC 2020), Globally online, November 1-6, 2020 (UTC) (CEUR Workshop Proceedings, Vol. 2721)*, Kerry L. Taylor, Rafael S. Gonçalves, Freddy Lécué, and Jun Yan (Eds.). CEUR-WS.org, 242–247. <http://ceur-ws.org/Vol-2721/paper561.pdf>